

Q1 - 25 January - Shift 1

Let $f : (0,1) \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \frac{1}{1-e^{-x}}, \text{ and}$$

$g(x) = (f(-x) - f(x))$. Consider two statements

(I) g is an increasing function in $(0, 1)$

(II) g is one-one in $(0, 1)$

Then,

(1) Only (I) is true

(2) Only (II) is true

(3) Neither (I) nor (II) is true

(4) Both (I) and (II) are true

Space for your notes:

Q2 - 25 January - Shift 1

Let $x = 2$ be a local minima of the function

$$f(x) = 2x^4 - 18x^2 + 8x + 12, \quad x \in (-4, 4).$$

If M is local maximum value of the function f in $(-4, 4)$,

then $M =$

(1) $12\sqrt{6} - \frac{33}{2}$ (2) $12\sqrt{6} - \frac{31}{2}$

(3) $18\sqrt{6} - \frac{33}{2}$ (4) $18\sqrt{6} - \frac{31}{2}$

Space for your notes:

Q3 - 25 January - Shift 2

Let the function $f(x) = 2x^3 + (2p-7)x^2 + 3(2p-9)x - 6$ have a maxima for some value of $x < 0$ and a minima for some value of $x > 0$. Then, the set of all values of p is

- (1) $\left(\frac{9}{2}, \infty\right)$ (2) $\left(0, \frac{9}{2}\right)$
 (3) $\left(-\infty, \frac{9}{2}\right)$ (4) $\left(-\frac{9}{2}, \frac{9}{2}\right)$

Space for your notes:

Q4 - 29 January - Shift 2

If the equation of the normal to the curve $y = \frac{x-a}{(x+b)(x-2)}$ at the point $(1, -3)$ is $x - 4y = 13$, then the value of $a + b$ is equal to _____.

Space for your notes:

Q5 - 29 January - Shift 2

Let $\alpha_1, \alpha_2, \dots, \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1| \geq |\alpha_2| \geq \dots \geq |\alpha_7|$. Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____.

Space for your notes:

Q6 - 30 January - Shift 1

The number of points on the curve $y = 54x^3 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is :

- (1) 2 (2) 3
(3) 4 (4) 0

Space for your notes:

Q7 - 30 January - Shift 2

#MathBoleTohMathonGo

If the functions $f(x) = \frac{x}{3} + 2bx + \frac{1}{2}$ and

$g(x) = \frac{x^3}{3} + ax + bx^2, a \neq 2b$ have a common

extreme point, then $a + 2b + 7$ is equal to

(1) 4

(2) $\frac{3}{2}$

(3) 3

(4) 6

Space for your notes:

Q8 - 31 January - Shift 1

A wire of length 20 m is to be cut into two pieces.

A piece of length ℓ_1 is bent to make a square of area A_1 and the other piece of length ℓ_2 is made

into a circle of area A_2 . If $2A_1 + 3A_2$ is minimum

then $(\pi \ell_1) : \ell_2$ is equal to:

(1) 6 : 1

(2) 3 : 1

(3) 1 : 6

(4) 4 : 1

Space for your notes:

Q9 - 01 February - Shift 1

Let $f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$,

$x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$. If α and β respectively are the maximum and the minimum values of f , then

(1) $\beta^2 - 2\sqrt{\alpha} = \frac{19}{4}$

(2) $\beta^2 + 2\sqrt{\alpha} = \frac{19}{4}$

(3) $\alpha^2 - \beta^2 = 4\sqrt{3}$

(4) $\alpha^2 + \beta^2 = \frac{9}{2}$

Space for your notes:

Q10 - 01 February - Shift 1

Let $f(x) = 2x + \tan^{-1}x$ and $g(x) = \log_e(\sqrt{1+x^2} + x)$, $x \in [0, 3]$. Then

(1) There exists $\hat{x} \in [0, 3]$ such that $f'(\hat{x}) < g'(\hat{x})$

(2) $\max f(x) > \max g(x)$

(3) There exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x)$,

$\forall x \in (x_1, x_2)$

(4) $\min f'(x) = 1 + \max g'(x)$

Space for your notes:

Q11 - 01 February - Shift 1

If $f(x) = x^2 + g'(1)x + g''(2)$ and

$g(x) = f(1)x^2 + xf'(x) + f''(x)$,

then the value of $f(4) - g(4)$ is equal to _____

Space for your notes:

Q12 - 01 February - Shift 1

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such

Space for your notes:

that $f'(x) + f(x) = \int_0^2 f(t) dt$. If $f(0) = e^{-2}$, then

$2f(0) - f(2)$ is equal to _____.

Q13 - 01 February - Shift 2

The sum of the absolute maximum and minimum values of the function $f(x) = |x^2 - 5x + 6| - 3x + 2$ in the interval $[-1, 3]$ is equal to :

Space for your notes:

(1) 10

(2) 12

(3) 13

(4) 24

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (4)**Q2 (1)****Q3 (3)****Q4 (4)****Q5 (9)****Q6 (3)****Q7 (4)****Q8 (1)****Q9 (1)****Q10 (2)****Q11 (14)****Q12 (1)****Q13 (1)**

#MathBoleTohMathonGo

Q1 (4)

$$g(x) = f(-x) - f(x) = \frac{1+e^x}{1-e^x}$$

$$\Rightarrow g'(x) = \frac{2e^x}{(1-e^x)^2} > 0$$

 $\Rightarrow g$ is increasing in $(0, 1)$ $\Rightarrow g$ is one-one in $(0, 1)$

Q2 (1)

$$f'(x) = 8x^3 - 36x + 8 = 4(2x^3 - 9x + 2)$$

$$f'(x) = 0$$

$$\therefore x = \frac{\sqrt{6}-2}{2}$$

Now

$$f(x) = \left(x^2 - 2x - \frac{9}{2}\right)(2x^2 + 4x - 1) + 24x + 7.5$$

$$\therefore f\left(\frac{\sqrt{6}-2}{2}\right) = M = 12\sqrt{6} - \frac{33}{2}$$

Q3 (3)

$$f(x) = 2x^3 + (2p - 7)x^2 + 3(2p - 9)x - 6$$

$$f'(x) = 6x^2 + 2(2p - 7)x + 3(2p - 9)$$

$$f'(0) < 0$$

$$\therefore 3(2p - 9) < 0$$

$$p < \frac{9}{2}$$

$$p \in \left(-\infty, \frac{9}{2}\right)$$

Q4 (4)

#MathBoleTohMathonGo

$$y = \frac{x - a}{(x + b)(x - 2)}$$

At point (1, -3),

$$-3 = \frac{1 - a}{(1 + b)(1 - 2)}$$

$$\Rightarrow 1 - a = 3(1 + b) \quad \dots (1)$$

$$\text{Now, } y = \frac{x - a}{(x + b)(x - 2)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(x + b)(x - 2) \times (1) - (x - a)(2x + b - 2)}{(x + b)^2 (x - 2)^2}$$

$$\text{At } (1, -3) \text{ slope of normal is } \frac{1}{4} \text{ hence } \frac{dy}{dx} = -4,$$

$$\text{So, } -4 = \frac{(1 + b)(-1) - (1 - a)b}{(1 + b)^2 (-1)^2}$$

Using equation (1)

$$\Rightarrow -4 = \frac{(1 + b)(-1) - 3(b + 1)b}{(1 + b)^2}$$

$$\Rightarrow -4 = \frac{(-1) - 3b}{(1 + b)} (b \neq -1)$$

$$\Rightarrow b = -3$$

$$\text{So, } a = 7$$

$$\text{Hence, } a + b = 7 - 3 = 4$$

Q5 (9)

Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____.

Given equation can be rearranged as

$$x(x^6 + 3x^4 - 13x^2 - 15) = 0$$

clearly $x = 0$ is one of the root and other part can

be observed by replacing $x^2 = t$ from which we

$$\text{have } t^3 + 3t^2 - 13t - 15 = 0$$

$$\Rightarrow (t - 3)(t^2 + 6t + 5) = 0$$

$$\text{So, } t = 3, t = -1, t = -5$$

Now we are getting $x^2 = 3, x^2 = -1, x^2 = -5$

$$\Rightarrow x = \pm\sqrt{3}, x = \pm i, x = \pm\sqrt{5}i$$

From the given condition $|\alpha_1| \geq |\alpha_2| \geq \dots \geq |\alpha_7|$

We can clearly say that $|\alpha_7| = 0$ and

$$\text{and } |\alpha_6| = \sqrt{5} = |\alpha_5|$$

$$\text{and } |\alpha_4| = \sqrt{3} = |\alpha_3| \text{ and } |\alpha_2| = 1 = |\alpha_1|$$

So we can have, $\alpha_1 = \sqrt{5}i, \alpha_2 = -\sqrt{5}i, \alpha_3 = \sqrt{3}i,$

$$\alpha_4 = -\sqrt{3}i, \alpha_5 = i, \alpha_6 = -i$$

Hence

$$\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$$

$$= 1 - (-3) + 5 = 9$$

Q6 (3)

Normal of line is parallel to line $x + 90y + 2 = 0$

$$m_N = -\frac{1}{90}$$

$$-\left(\frac{dx}{dy}\right)_{(x_1, y_1)} = -\frac{1}{90} \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = 90$$

Now,

$$\frac{dy}{dx} = 270x^4 - 540x^3 - 210x^2 + 360x + 210 = 90$$

$$\Rightarrow x = 1, 2, \frac{-2}{3}, \frac{-1}{3}$$

(4) normals

Q7 (4)

$$f'(x) = x^2 + 2b + ax$$

$$g'(x) = x^2 + a + 2bx$$

$$(2b - a) - x(2b - a) = 0$$

$\therefore x = 1$ is the common root

Put $x = 1$ in $f'(x) = 0$ or $g'(x) = 0$

$$1 + 2b + a = 0$$

$$7 + 2b + a = 6$$

Q8 (1)

$$\ell_1 + \ell_2 = 20 \Rightarrow \frac{d\ell_2}{d\ell_1} = -1$$

$$A_1 = \left(\frac{\ell_1}{4}\right)^2 \text{ and } A_2 = \pi \left(\frac{\ell_2}{2\pi}\right)^2$$

$$\text{Let } S = 2A_1 + 3A_2 = \frac{\ell_1^2}{8} + \frac{3\ell_2^2}{4\pi}$$

$$\frac{ds}{d\ell} = 0 \Rightarrow \frac{2\ell_1}{8} + \frac{6\ell_2}{4\pi} \cdot \frac{d\ell_2}{d\ell_1} = 0$$

$$\Rightarrow \frac{\ell_1}{4} = \frac{6\ell_2}{4\pi} \Rightarrow \frac{\pi\ell_1}{\ell_2} = 6$$

Q9 (1)

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$f(x) = \begin{vmatrix} 2 + \sin 2x & \cos^2 x & \sin 2x \\ 2 + \sin 2x & 1 + \cos^2 x & \sin 2x \\ 2 + \sin 2x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$$

$$f(x) = (2 + \sin 2x) \begin{vmatrix} 1 & \cos^2 x & \sin 2x \\ 1 & 1 + \cos^2 x & \sin 2x \\ 1 & \cos^2 x & 1 + \sin 2x \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$f(x) = (2 + \sin 2x) \begin{vmatrix} 1 & \cos^2 x & \sin 2x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= (2 + \sin 2x) (1) = 2 + \sin 2x$$

$$= \sin 2x \in \left[\frac{\sqrt{3}}{2}, 1 \right]$$

$$\text{Hence } 2 + \sin 2x \in \left[2 + \frac{\sqrt{3}}{2}, 3 \right]$$

Q10 (2)

$$f(x) = 2x + \tan^{-1} x \text{ and } g(x) = \ln(\sqrt{1+x^2} + x)$$

$$\text{and } x \in [0, 3]$$

$$g'(x) = \frac{1}{\sqrt{1+x^2}}$$

$$\text{Now, } 0 \leq x \leq 3$$

$$0 \leq x^2 \leq 9$$

$$1 \leq 1+x^2 \leq 10$$

$$\text{So, } 2 + \frac{1}{10} \leq f'(x) \leq 3$$

$$\frac{21}{10} \leq f'(x) \leq 3 \text{ and } \frac{1}{\sqrt{10}} \leq g'(x) \leq 1$$

option (4) is incorrect

$$\text{From above, } g'(x) < f'(x) \forall x \in [0, 3]$$

Option (1) is incorrect.

$f'(x)$ & $g'(x)$ both positive so $f(x)$ & $g(x)$ both are

increasing

So,

$$\max(f(x) \text{ at } x=3 \text{ is } 6 + \tan^{-1} 3$$

$$\text{Max } (g(x) \text{ at } x=3 \text{ is } \ln(3 + \sqrt{10}))$$

$$\text{And } 6 + \tan^{-1} 3 > \ln(3 + \sqrt{10})$$

Option (2) is correct

Q11 (14)

$$f(x) = x^2 + g'(1)x + g''(2)$$

$$f'(x) = 2x + g'(1)$$

$$f''(x) = 2$$

$$g(x) = f(1)x^2 + x[2x + g'(1)] + 2$$

$$g'(x) = 2f(1)x + 4x + g'(1)$$

$$g''(x) = 2f(1) + 4$$

$$g''(x) = 0$$

$$2f(1) + 4 = 0$$

$$f(1) = -2$$

$$-2 = 1 + g'(1) = g'(1) = -3$$

$$\text{So, } f'(x) = 2x - 3$$

$$f(x) = x^2 - 3x + c$$

$$c = 0$$

$$f(x) = x^2 - 3x$$

$$g(x) = -3x + 2$$

$$f(4) - g(4) = 14$$

Q12 (1)

$$\frac{dy}{dx} + y = k$$

$$y \cdot e^x = k \cdot e^x + c$$

$$f(0) = e^{-2}$$

$$\Rightarrow c = e^{-2} - k$$

$$\therefore y = k + (e^{-2} - k)e^{-x}$$

$$\text{now } k = \int_0^2 (k + (e^{-2} - k)e^{-x}) dx$$

$$\Rightarrow k = e^{-2} - 1$$

$$\therefore y = (e^{-2} - 1) + e^{-x}$$

$$f(2) = 2e^{-2} - 1, f(0) = e^{-2}$$

$$2f(0) - f(2) = 1$$

Q13 (1)

$$f(x) = |x^2 - 5x + 6| - 3x + 2$$

$$f(x) = \begin{cases} x^2 - 8x + 8 & ; x \in [-1, 2] \\ -x^2 + 2x - 4 & ; x \in [2, 3] \end{cases}$$

